Revisiting Locally Differentially Private Protocols: Towards Better Trade-offs in Privacy, Utility, and Attack Resistance [EAB]

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* Work done while at Inria, France

Context & Motivation

Local Differential Privacy (LDP) [KLNRS11]

Definition (Local Differential Privacy)

Let $\epsilon > 0$, a randomized mechanism $\mathcal{M}$ satisfies ϵ-local differential privacy (ϵ-LDP), if for any two inputs $v, v' \in \text{Domain}(\mathcal{M})$ and for any output $z \in \text{Range}(\mathcal{M})$

$$\frac{\Pr[\mathcal{M}(v) = z]}{\Pr[\mathcal{M}(v') = z]} \leq e^{\epsilon}$$

Privacy loss

[KLNRS11] What can we learn privately?. SIAM Journal on Computing 2011.

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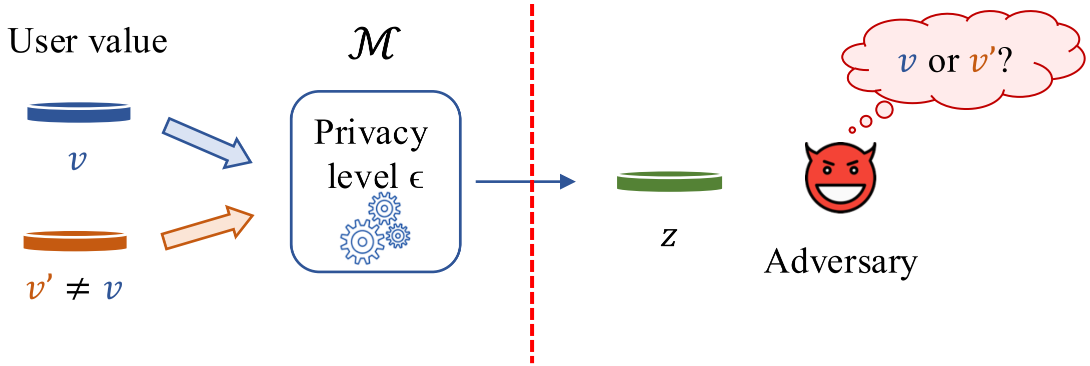
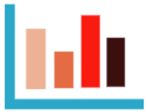


Any output should be about as likely regardless of the input value

[KLNRS11] What can we learn privately?. SIAM Journal on Computing 2011.

Frequency Estimation Under LDP

Assumption: Each user holds a value v from a a finite domain V (size $k = |V|$)



Goal: Estimate the frequency (or histogram) of any value $v \in V$

General Scheme for Frequency Estimation Under LDP

Input: Original data of users, privacy parameter ϵ, and LDP protocol $\mathcal{M}$

Output: k-bins histogram

 # User-side

for each user i with input value $v_i \in V$ **do**

$x_i = \textbf{Encode}(v_i)$ (if needed)

$y_i = \textbf{Perturb}(x_i)$ with $\mathcal{M}$

Transmit y_i to the server

 # Server-side

The server **Aggregates** the reported values and estimate their frequency

Real-World Deployments of LDP

Frequency ("monitoring") estimation based on LDP is (or has been) widely deployed!

- In Google Chrome browser, to collect browsing statistics (now deprecated) [EPK14]
- In Microsoft Windows, to collect telemetry data over time [DKS17]
- In Apple iOS and MacOS, to collect typing statistics [A17]
- In Google Gboard, for out-of-vocabulary word discovery [SKSGS24]

This yields deployments of over more than 100 million users…

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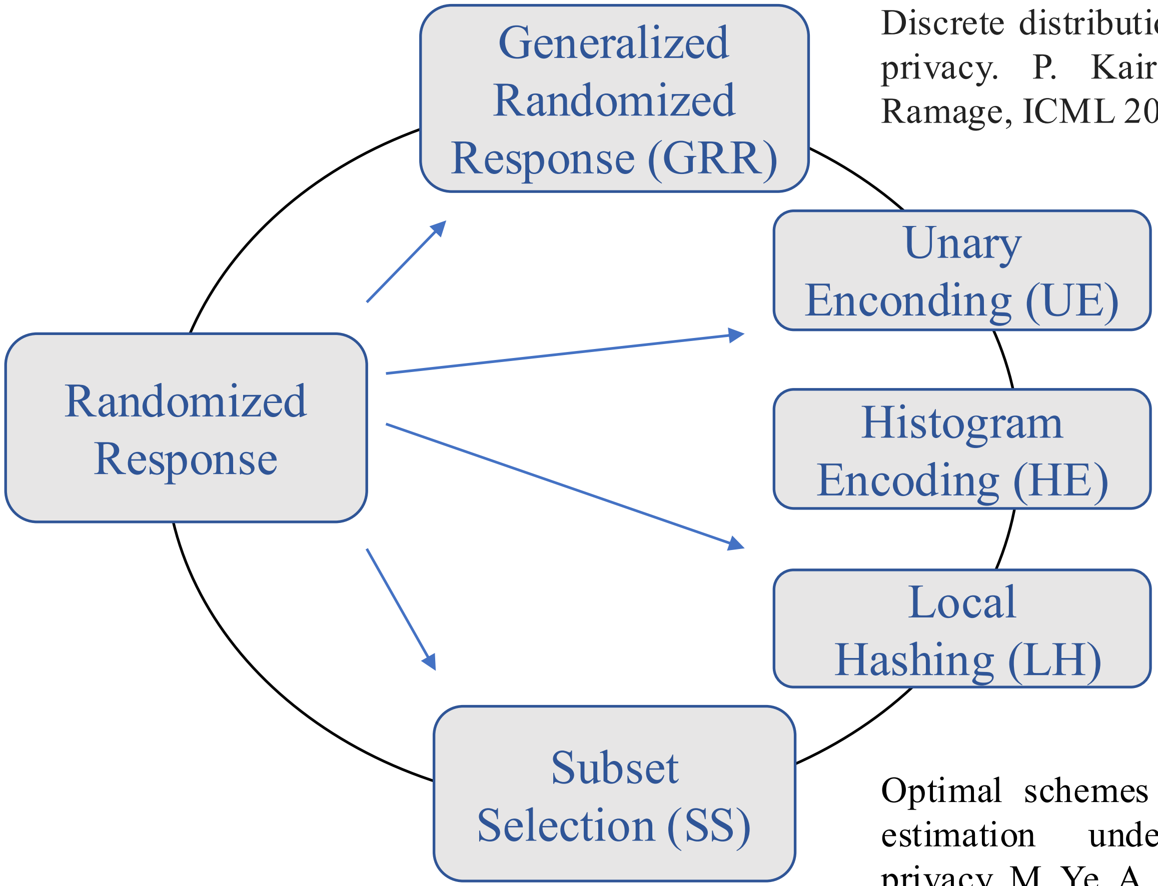
[EPK14] RAPPOR: Randomized Aggregatable Privacy-Preserving Ordinal Response. CCS 2014.
[DKS17] Collecting Telemetry Data Privately. NeurIPS 2017.
[A17] Learning with privacy at scale. White Paper 2017.
[SKSGS24] Private federated discovery of out-of-vocabulary words for gboard. Arxiv 2024.

Utility-Driven Design of LDP

From Two to Many Categories: State-of-the-Art LDP Protocols



Discrete distribution estimation under local privacy. P. Kairouz, K. Bonawitz, D. Ramage, ICML 2016

RAPPOR: Randomized Aggregatable Privacy-Preserving Ordinal Response. Ú. Erlingsson, V. Pihur, A. Korolova, CCS 2014

Locally Differentially Private Protocols for Frequency Estimation T. Wang, J. Blocki, N. Li, S. Jha: USENIX Security 2017

Optimal schemes for discrete distribution estimation under locally differential privacy. M. Ye, A. Barg: IEEE TIT 2018

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Utility-Driven Optimization in LDP

Most LDP frequency estimation protocols share the same unbiased estimator $\hat{f}(v)$

- Closed-form equation for variance

$$\mathrm{Var}^*[\hat{f}(v)] = \frac{q(1-q)}{n(p-q)^2} + \frac{f(v)(1-p-q)}{n(p-q)}$$

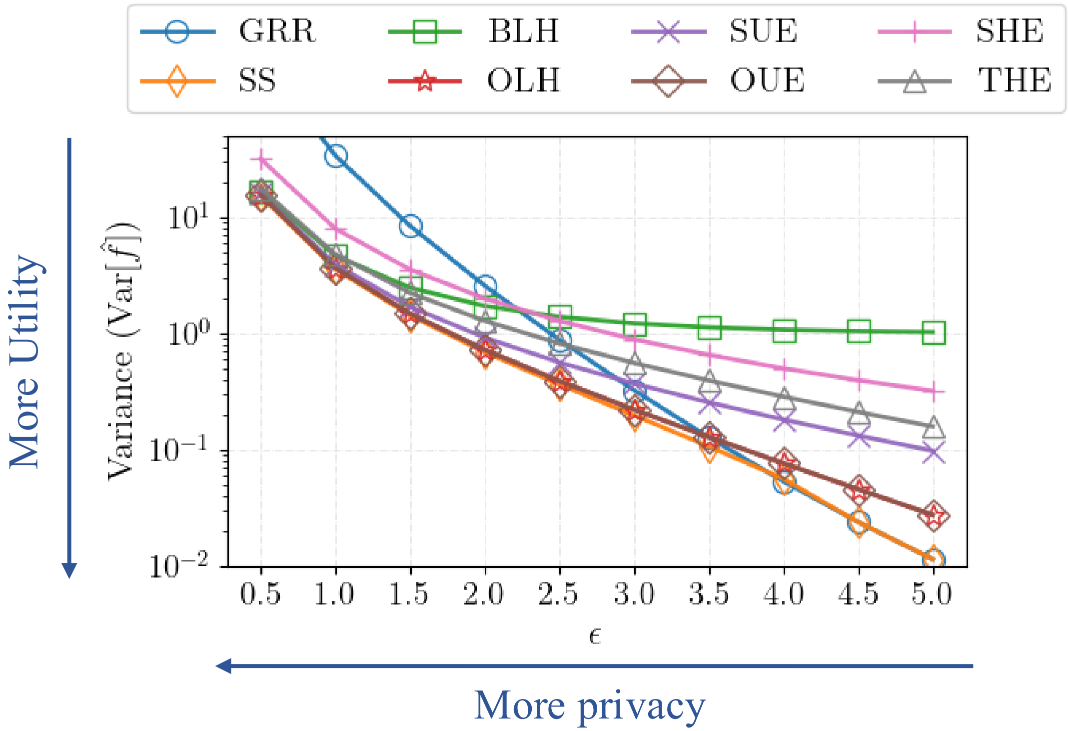
- Since unbiased → $\mathrm{Var}[\hat{f}(v)]$ = MSE (Mean Squared Error)

- Closed-form MSE enabled **utility-driven optimization** of LDP protocols [WBLJ17]

    - E.g., "Optimized UE (OUE)", "Optimized LH (OLH)"

[WBLJ17] Locally Differentially Private Protocols for Frequency Estimation. USENIX Security 2017.

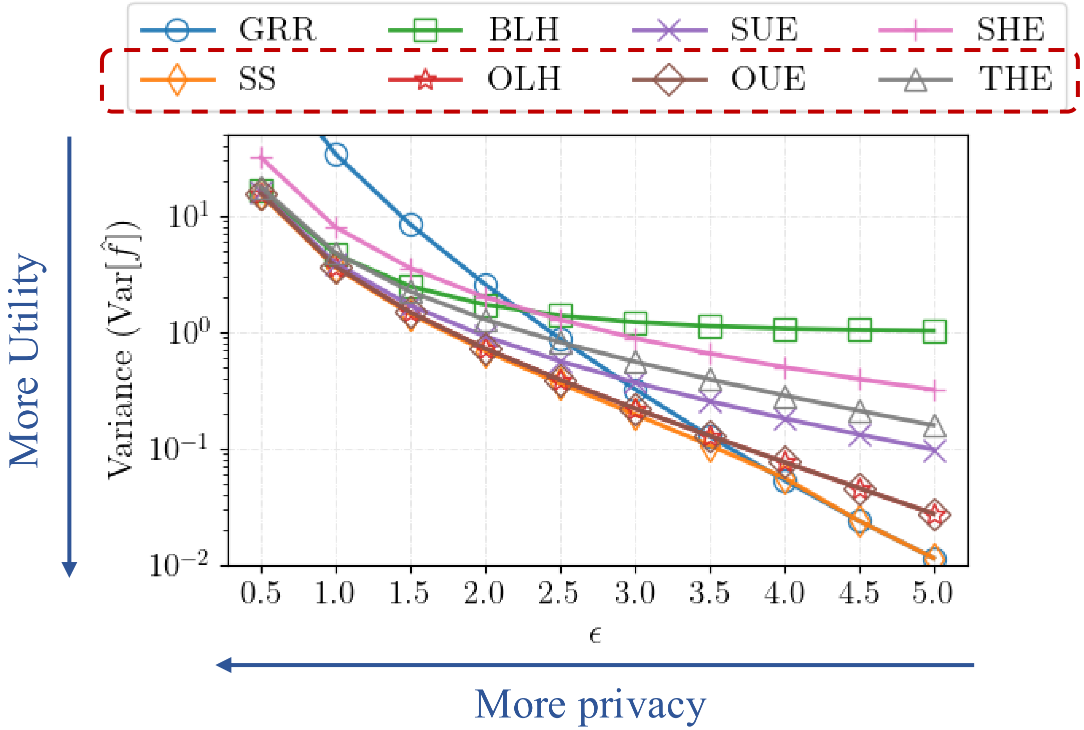
Privacy-Utility Trade-Off Across LDP Protocols



Privacy-Utility Trade-Off Across LDP Protocols

Protocols with same ϵ have very different utility

SS/OUE/OLH/THE optimized variance/MSE

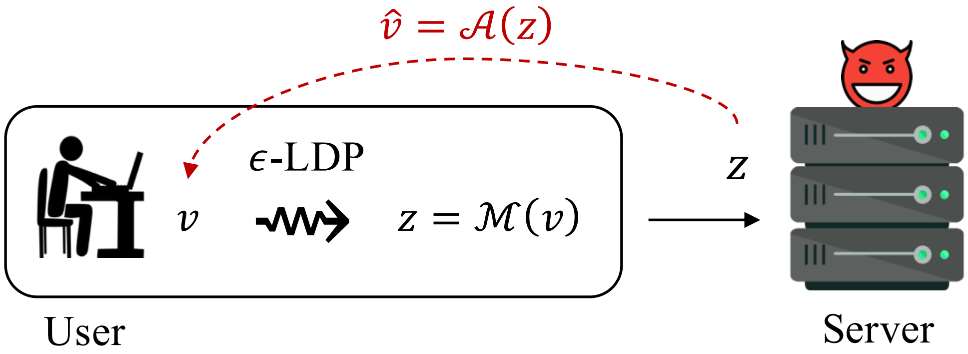


Adversarial Perspective on LDP

Data Reconstruction on LDP Protocols [GLCTW22, AGCP23]

ϵ-LDP is abstract $\rightarrow$ What does "$\epsilon = 2$" really mean for an individual?

- **Data reconstruction attack** gives a concrete interpretation
- Given $z = \mathcal{M}(v)$, adversary predicts the original value v



[GLCTW22] An Adversarial Approach to Protocol Analysis and Selection in LDP. TIFS 2022.
[AGCP23] On the Risks of Collecting Multidimensional Data Under LDP. VLDB 2023.

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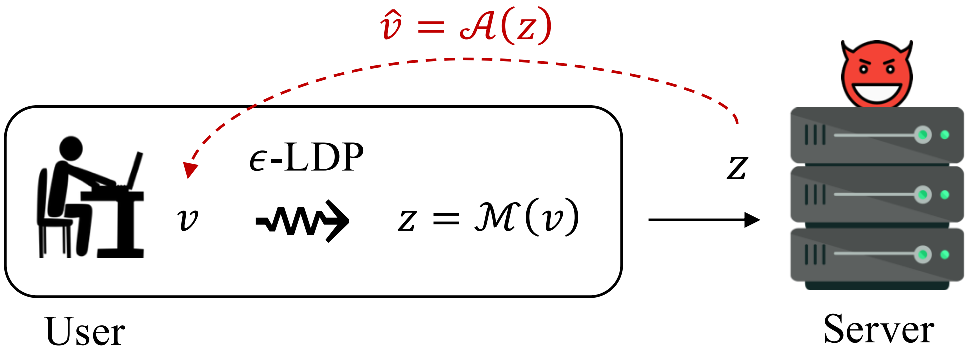
- Given $z = \mathcal{M}(v)$, adversary predicts the original value v
- Bayesian adversary → $\hat{v} = \underset{v \in V}{\operatorname{argmax}} \Pr[z \mid v] \cdot \Pr[v]$
- Metric: Adversarial Success Rate ($\mathrm{ASR} = \Pr[v = \hat{v}]$)
    - "If ASR $=$ 0.35, then the attacker has a 35% chance of reconstructing the true value"



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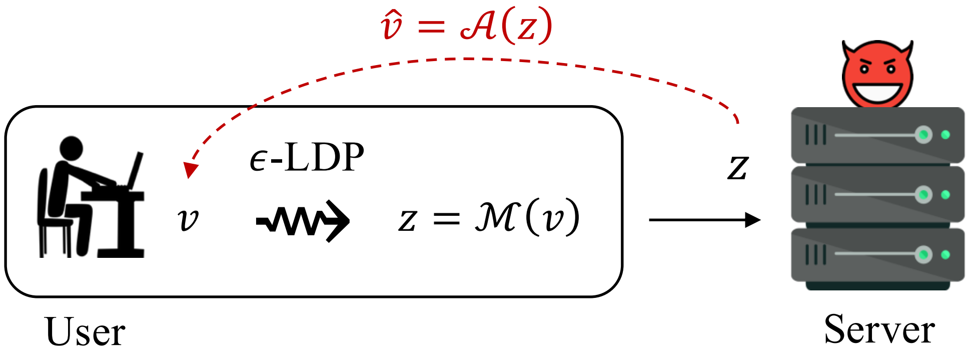
Uniform prior assumed
$\Pr[v] = \frac{1}{k}$



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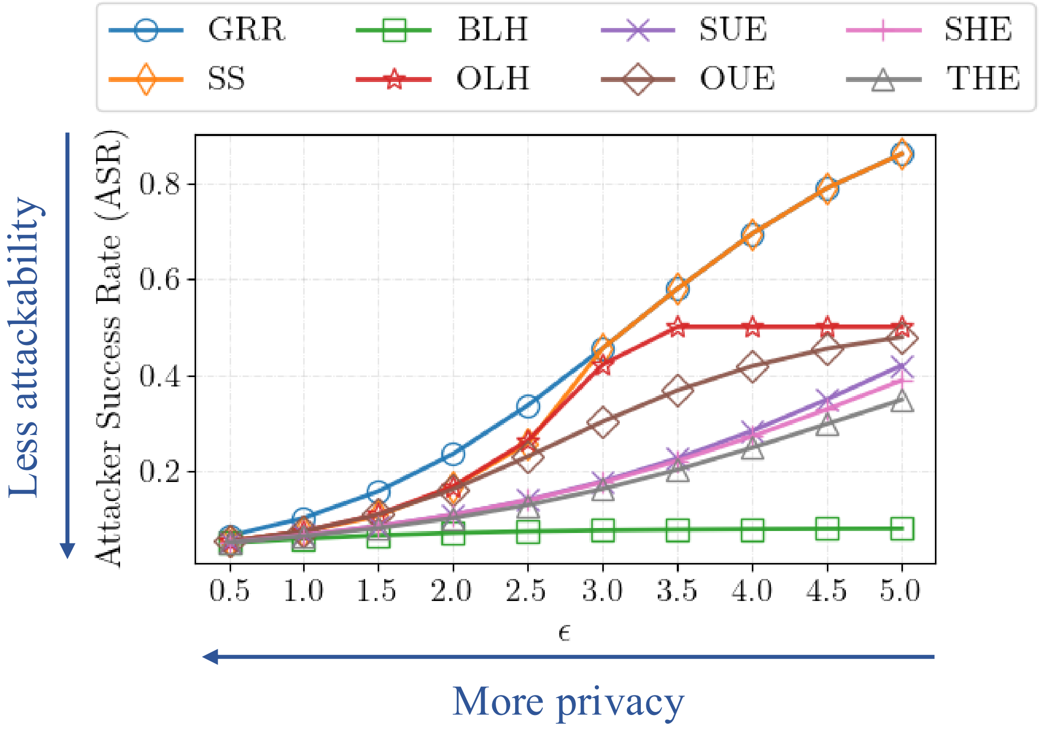
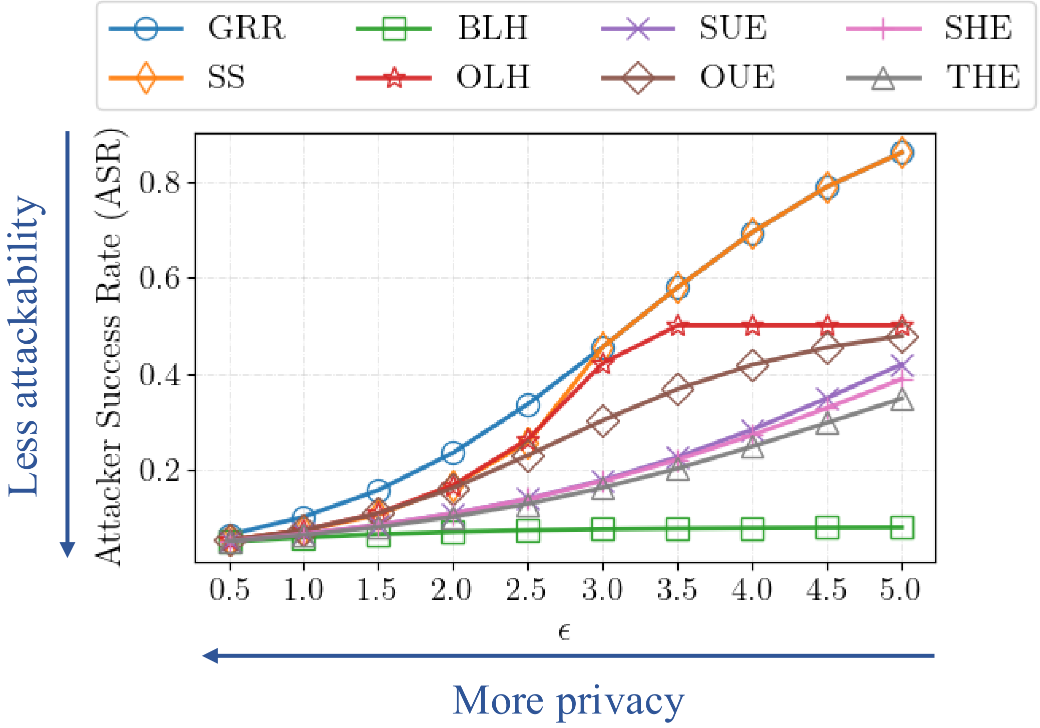
Reconstruction Rate Across LDP Protocols



Reconstruction Rate Across LDP Protocols

Protocols with same ϵ have very different vulnerabilities

No single protocol dominates → inherent trade-offs



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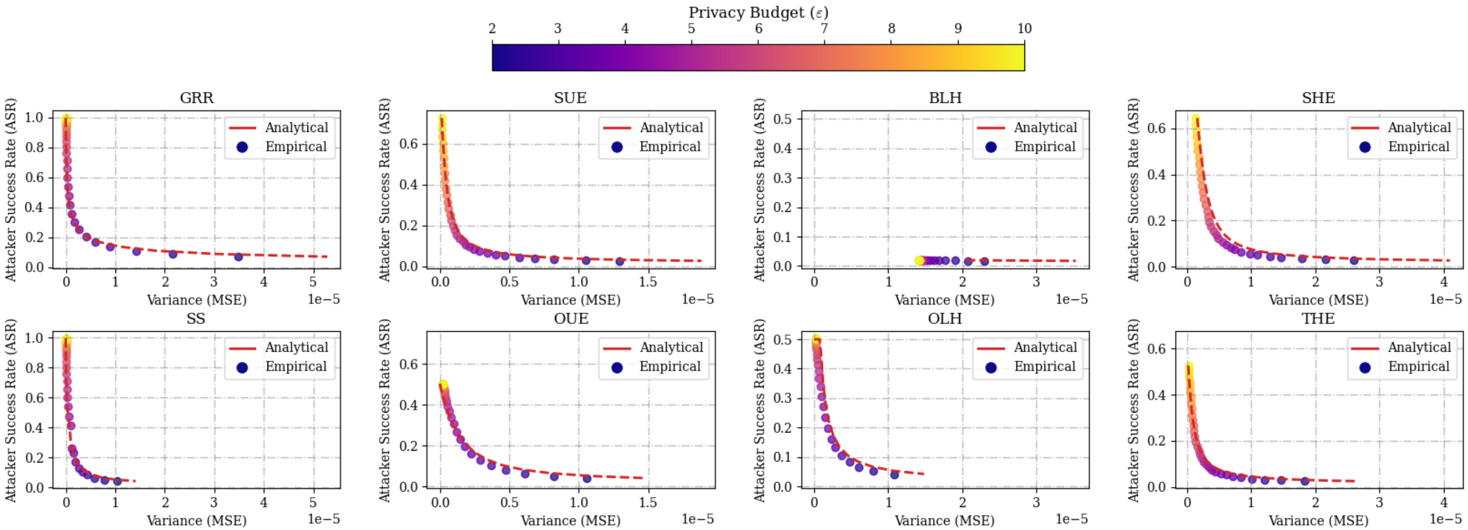
Our Proposed Joint ASR–MSE Analysis

From Trade-offs to Pareto Frontiers

Given ϵ, we obtain two axes from closed-form expressions

- **Utility loss (MSE):** how far estimates deviate
- **Privacy risk (ASR):** reconstruction attack

Key insight: These define a Pareto frontier → best achievable trade-offs



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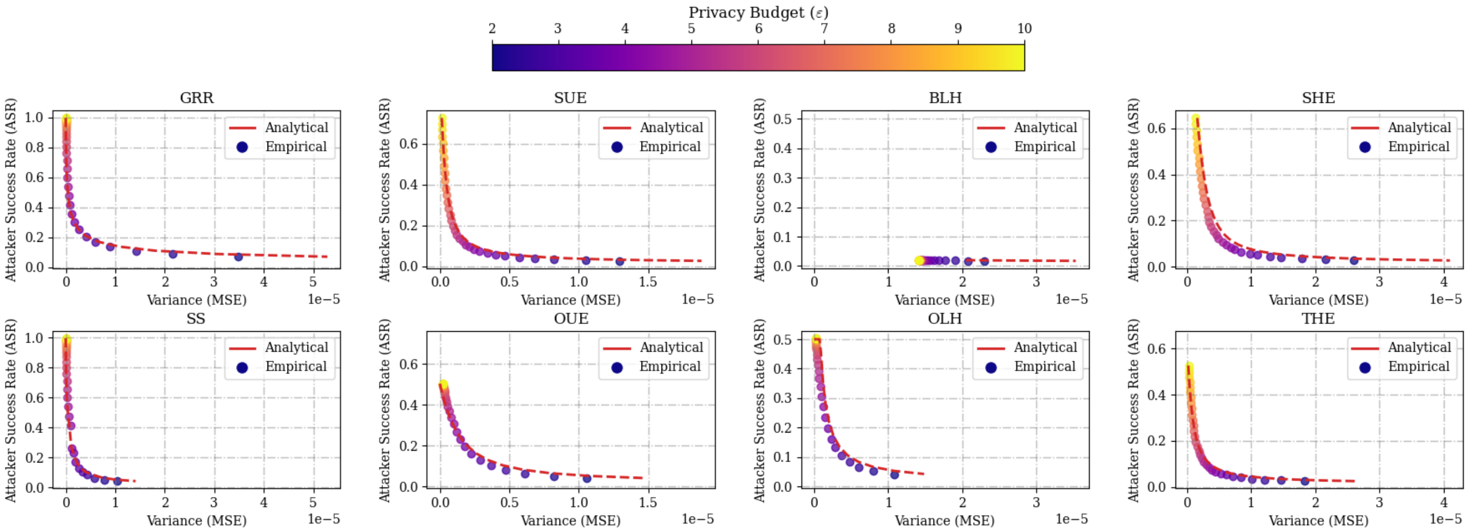


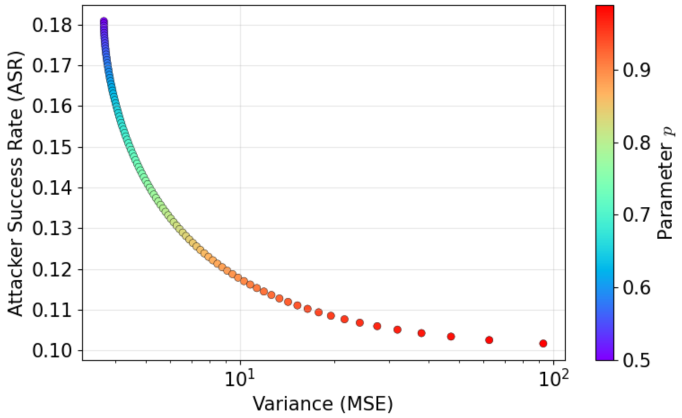
Empirical results **match** analytical curves

Joint Optimization of LDP Protocols (ASR–MSE)

Insight: Some protocols have tunable parameters → we can re-optimize them analytically

Our approach (for an LDP protocol parameterized by p)

- **Step 1:** Compute the Pareto frontier (ASR *vs.* MSE)



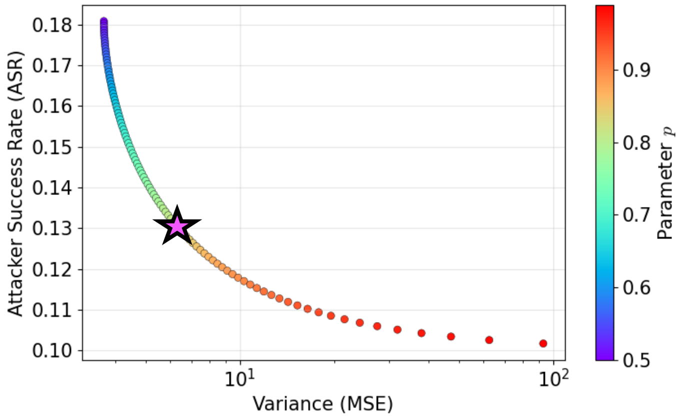
[DF07] Multi-Objective Optimization in Computer Networks Using Metaheuristics. Auerbach Publications 2007.

Joint Optimization of LDP Protocols (ASR–MSE)

Insight: Some protocols have tunable parameters → we can re-optimize them analytically

Our approach (for an LDP protocol parameterized by p)

- **Step 1:** Compute the Pareto frontier (ASR *vs.* MSE)
- **Step 2:** Select a point ★ with a multi-objective rule [DF07]
  - Weighted scalarization: $\min_{p} \quad J(p) = w_{\mathrm{ASR}} \cdot \mathrm{ASR} + w_{\mathrm{MSE}} \cdot \mathrm{MSE}$

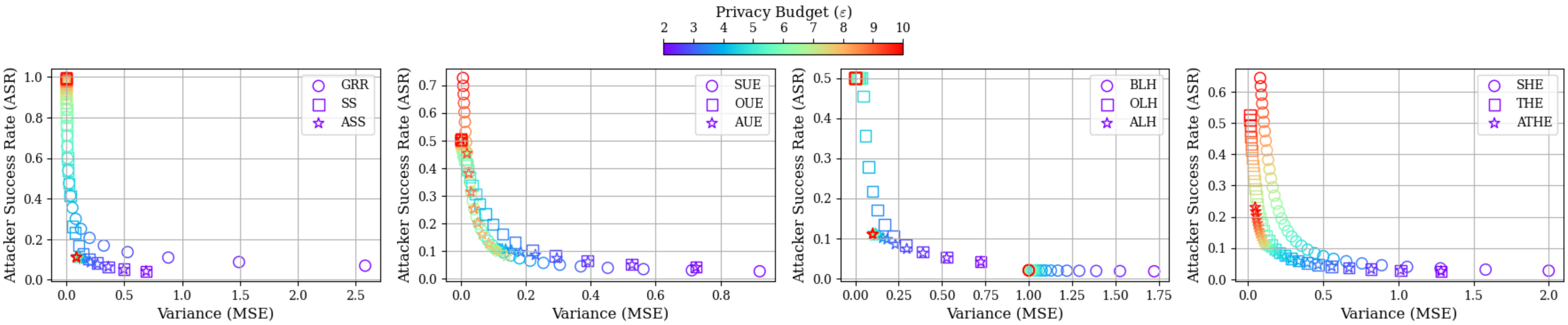


Protocol design becomes a controllable trade-off, not a fixed choice

[DF07] Multi-Objective Optimization in Computer Networks Using Metaheuristics. Auerbach Publications 2007.

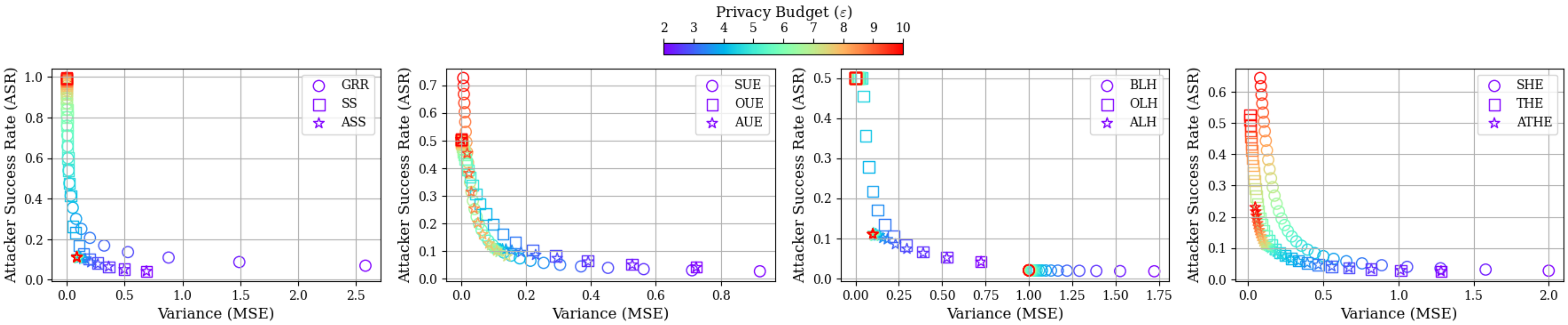
Adaptive LDP Protocols: Improved ASR–MSE Trade-offs

- Standard protocols become **adaptive** via parameter tuning (**ASS, AUE, ALH, ATHE**)



Adaptive LDP Protocols: Improved ASR–MSE Trade-offs

- Standard protocols become **adaptive** via parameter tuning (**ASS, AUE, ALH, ATHE**)
- Our adaptive protocols strictly dominate standard protocols on the Pareto frontier



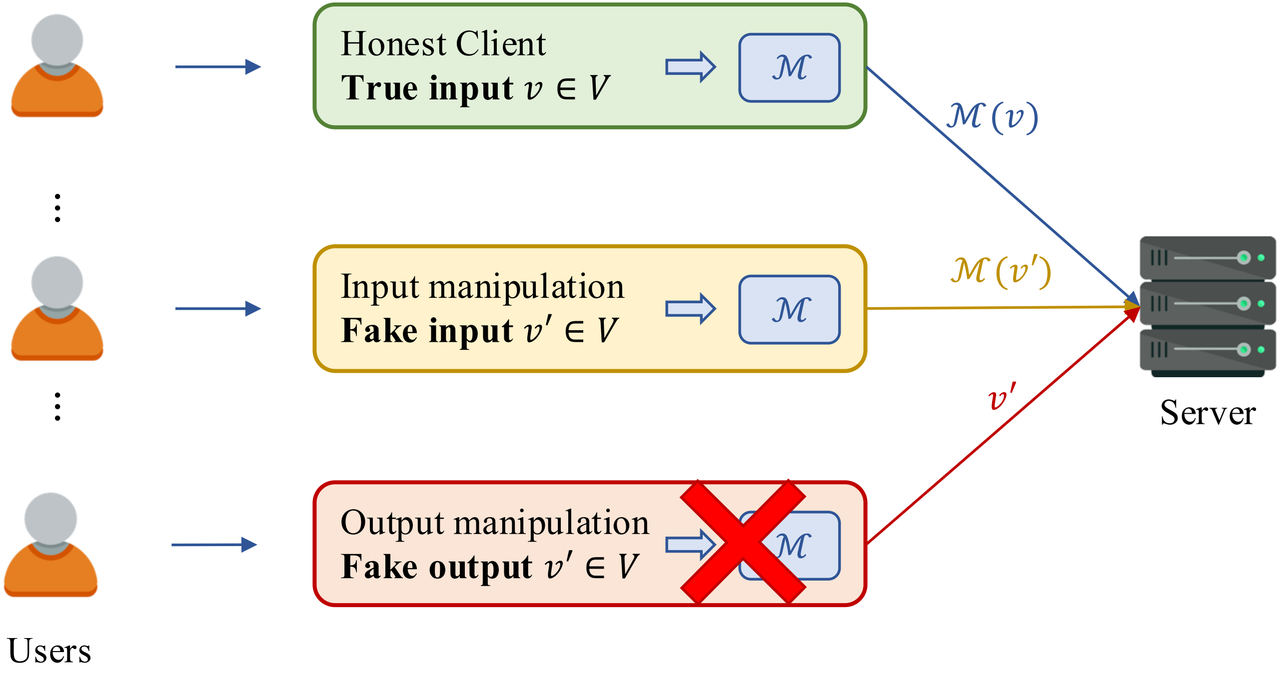
More experiments in the paper...

Beyond Privacy:
Adding Integrity

Extending the Trade-off: Adding Integrity Risk

Poisoning attacks [CJG21]

- Malicious users can manipulate inputs → Bias estimated statistics

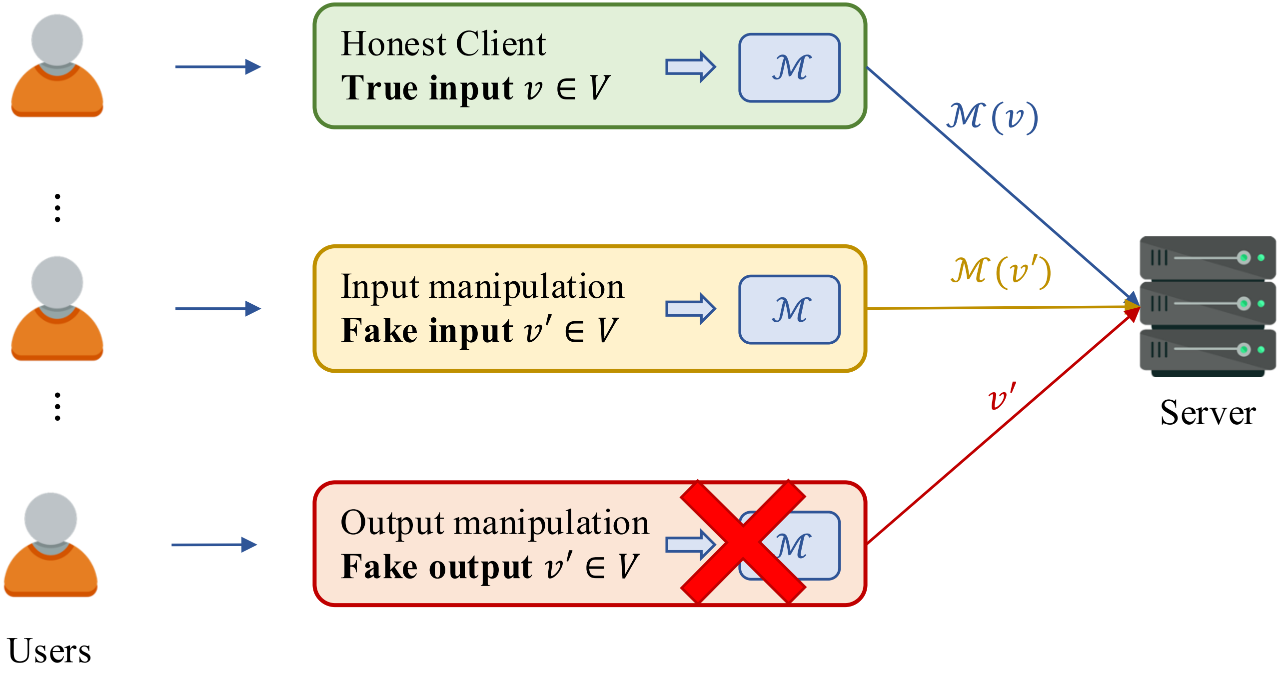


[CJG21] Data poisoning attacks to local differential privacy protocols. USENIX Security 2021.

Extending the Trade-off: Adding Integrity Risk

Poisoning attacks [CJG21]

- Malicious users can manipulate inputs → Bias estimated statistics
- For some protocols, the Expected Poisoning Gain (EPG) can be computed analytically



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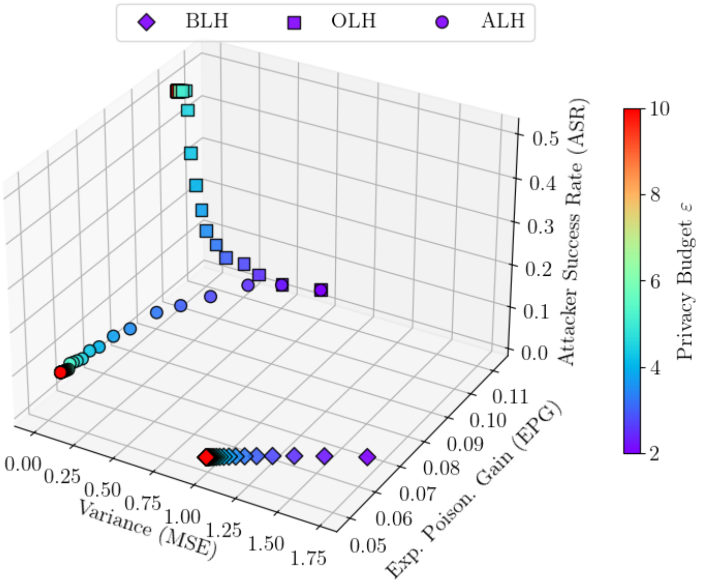
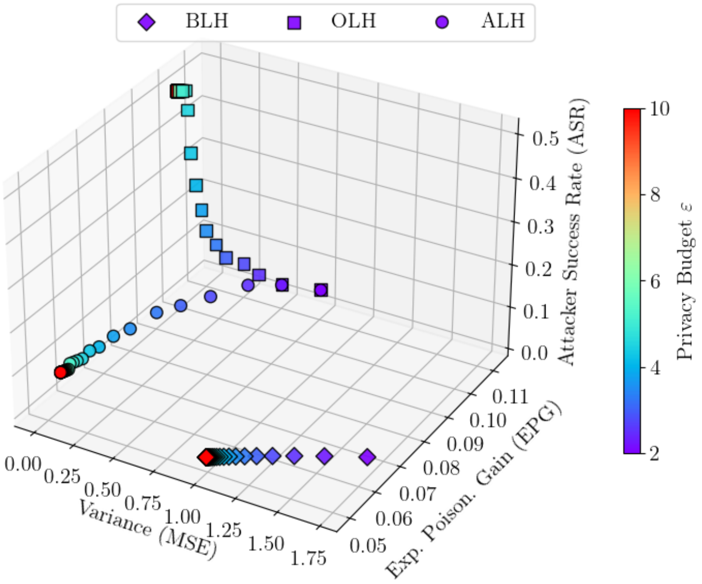
A Three-Way Trade-off: Privacy–Utility–Integrity



Here, we optimize ASR–MSE and **evaluate** the impact on EPG

A Three-Way Trade-off: Privacy–Utility–Integrity

- Our adaptive protocols also reduce EPG (even without optimizing for it)
- $\downarrow\epsilon \Rightarrow \downarrow$ASR but $\uparrow$MSE and $\uparrow$EPG



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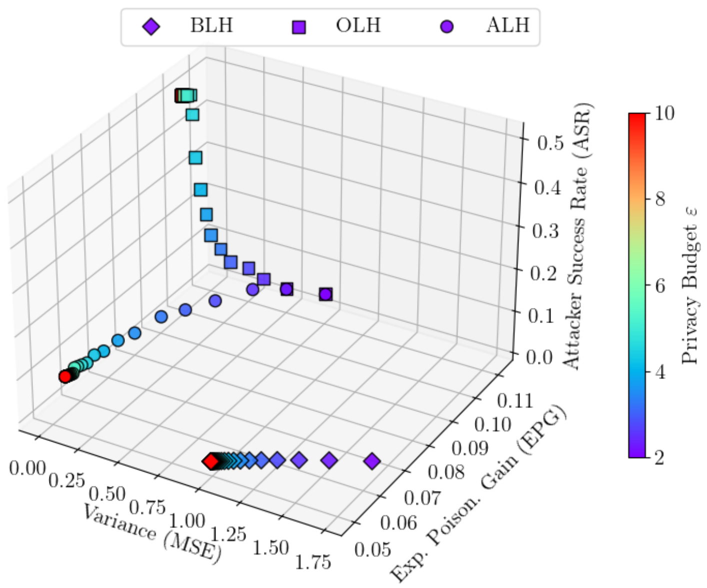
A Three-Way Trade-off: Privacy–Utility–Integrity

- Our adaptive protocols also reduce EPG (even without optimizing for it)
- $\downarrow\epsilon \Rightarrow \downarrow$ASR but $\uparrow$MSE and $\uparrow$EPG
- Our framework naturally extends to other risks (e.g., poisoning) → Three-way trade-off
  - E.g., "$w_{\text{ASR}} \cdot \text{ASR} + w_{\text{MSE}} \cdot \text{MSE} + w_{\text{EPG}} \cdot \text{EPG}$ "



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Conclusion

Final Remarks

Optimizing LDP only for utility is **not enough**

Designing LDP → optimizing risk–utility trade-offs

Key Takeaways

- Adversarial perspective → auditing, not just "breaking"
- ϵ alone is insufficient → measure real privacy risks
- Designing LDP = joint optimization: Privacy (ϵ) × Utility (MSE) × Attackability (ASR) × ...

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Perspectives

- Extending the optimization dimensions (e.g., integrity, communication cost)
- Richer adversarial analyses (e.g., correlation [JUDT25], longitudinal, adaptive attackers)
- Toward practical optimization tools (e.g., LDP-Toolbox [ZMA25])

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[JUDT25] The Hidden Cost of Correlation: Rethinking Privacy Leakage in Local Differential Privacy. ArXiv 2025.
[ZMA25] Demo: Exploring Utility and Attackability Trade-offs in Local Differential Privacy. CCS 2025.

Thank You! Questions?

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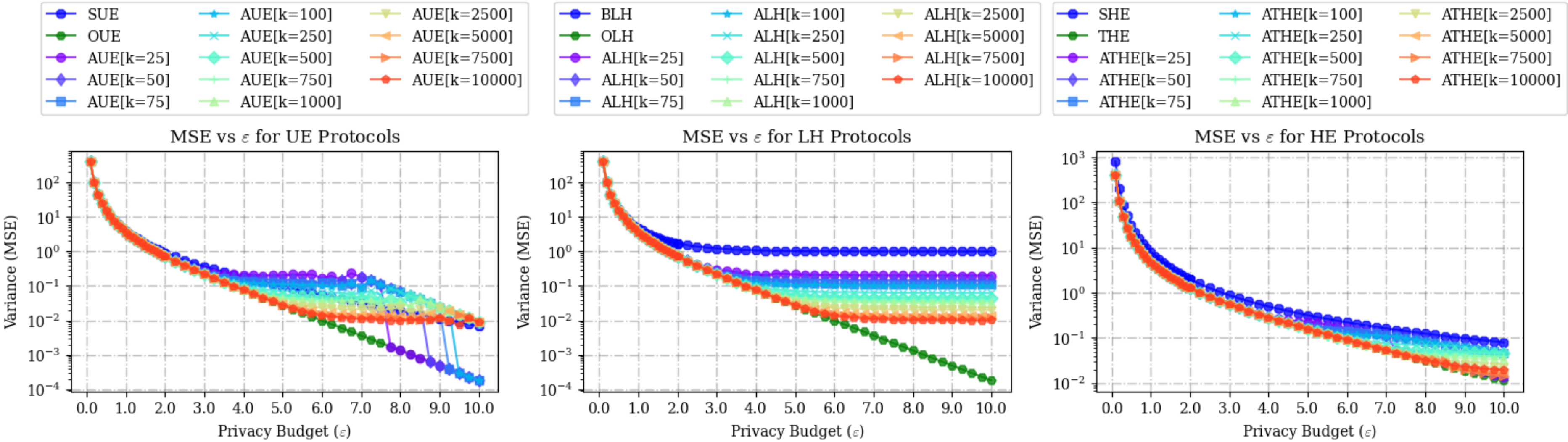
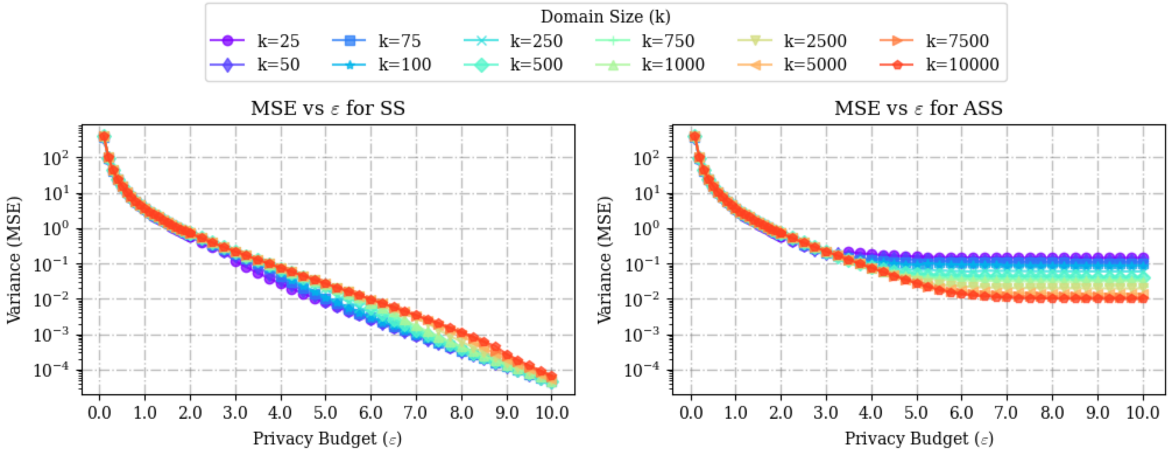
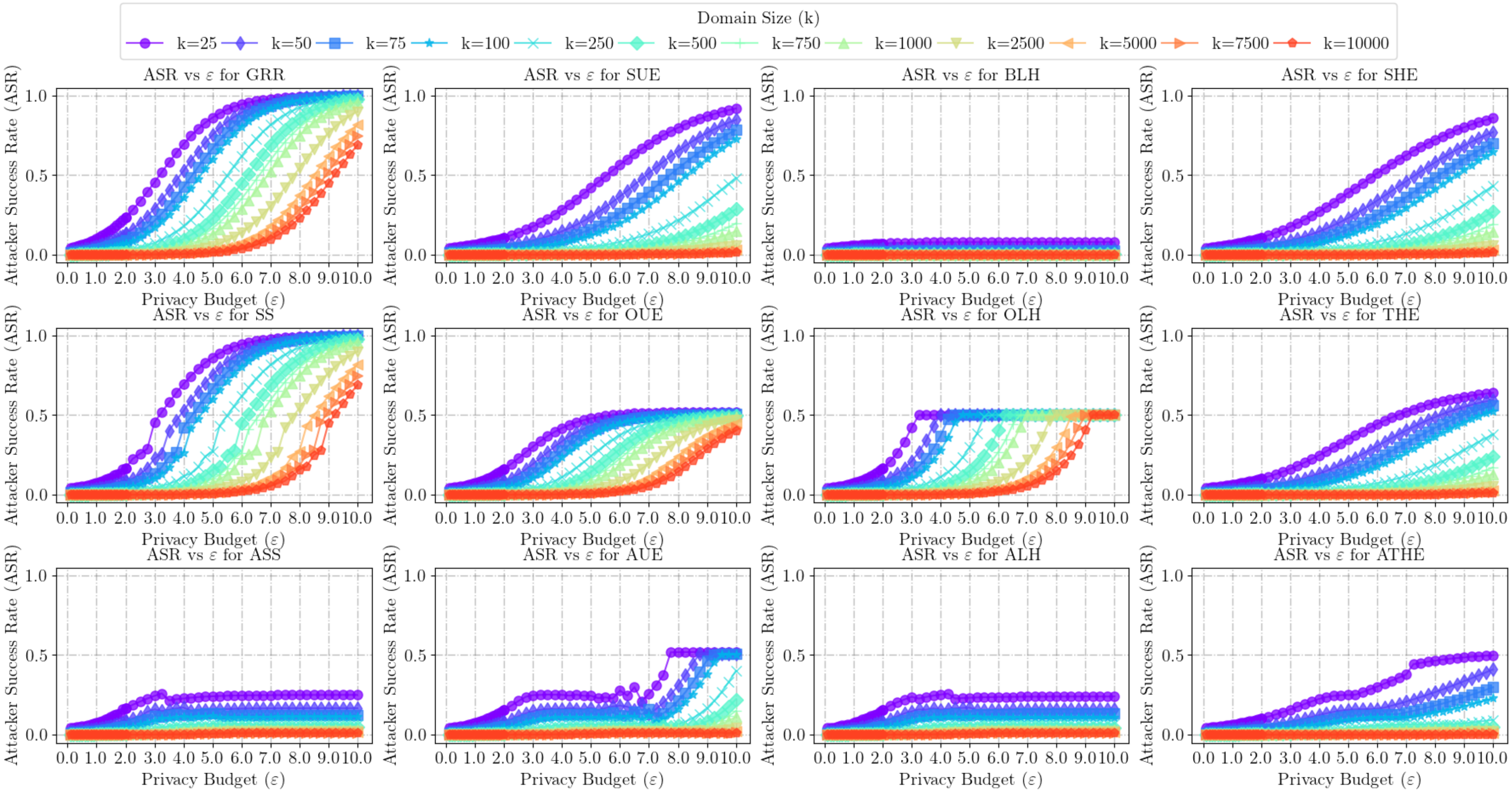
Appendix Slides

Privacy-Utility Trade-Offs

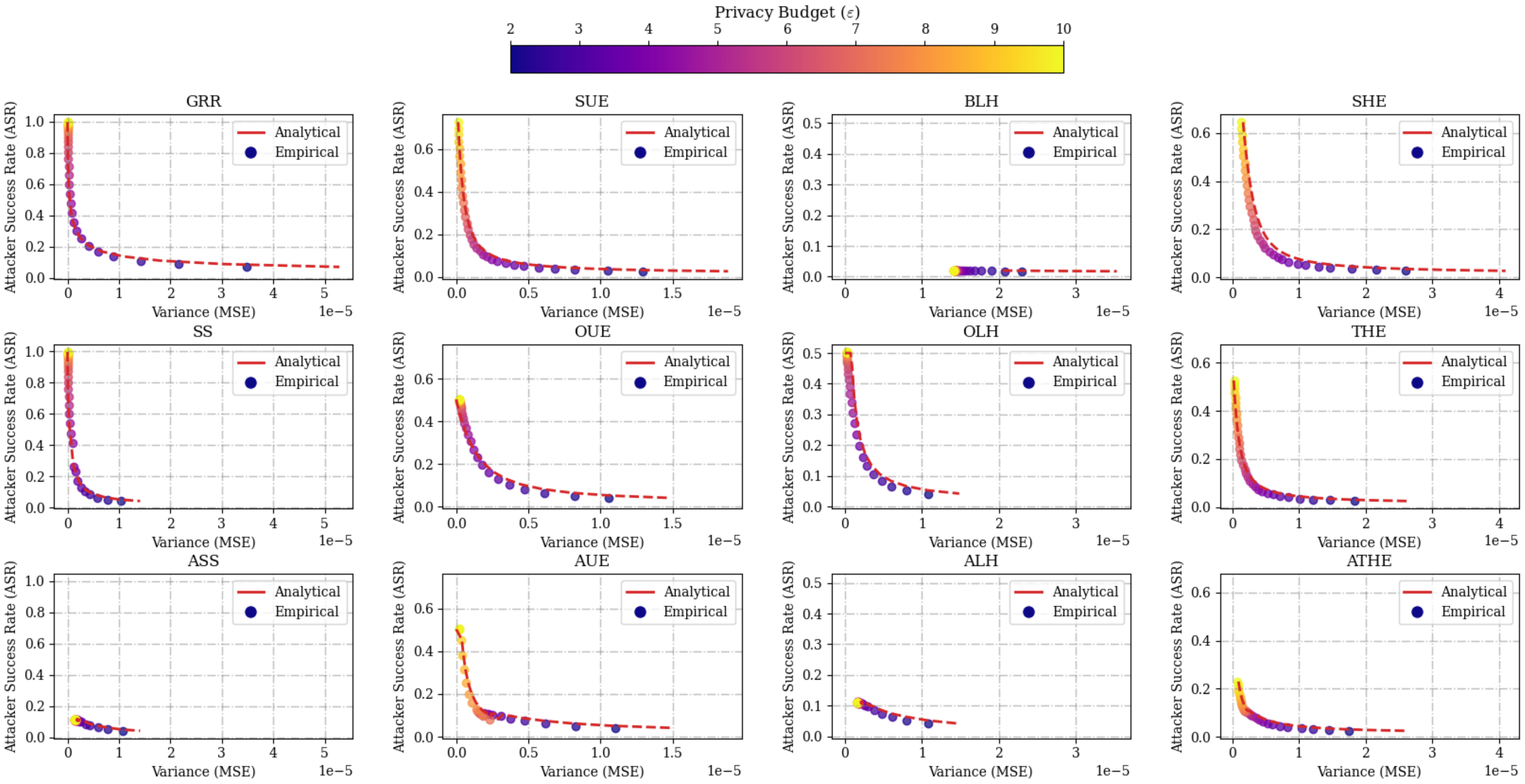




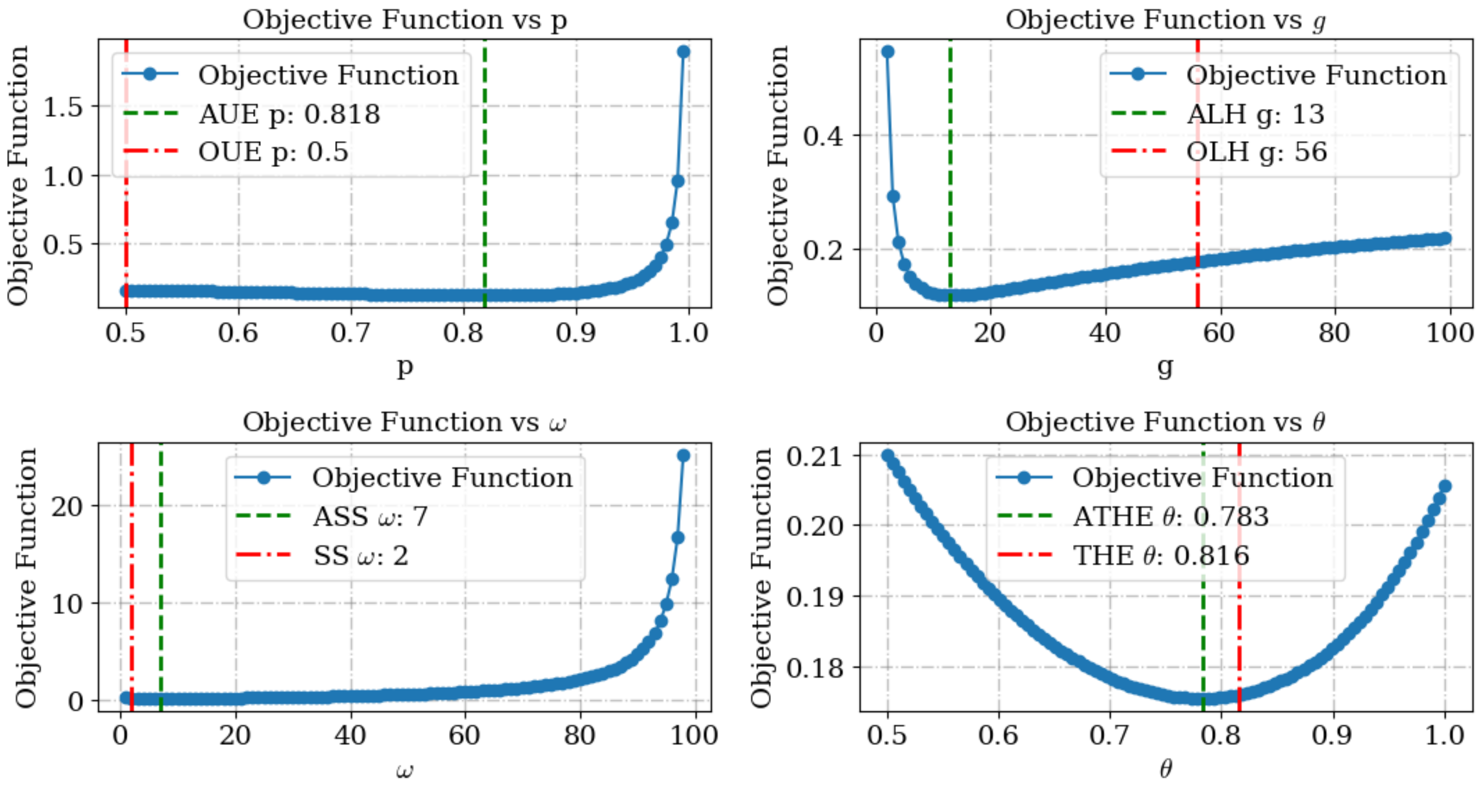
Privacy-Attackability Trade-Offs



Empirical *vs.* Analytical Pareto Frontiers



Parameter *vs.* Objective Function



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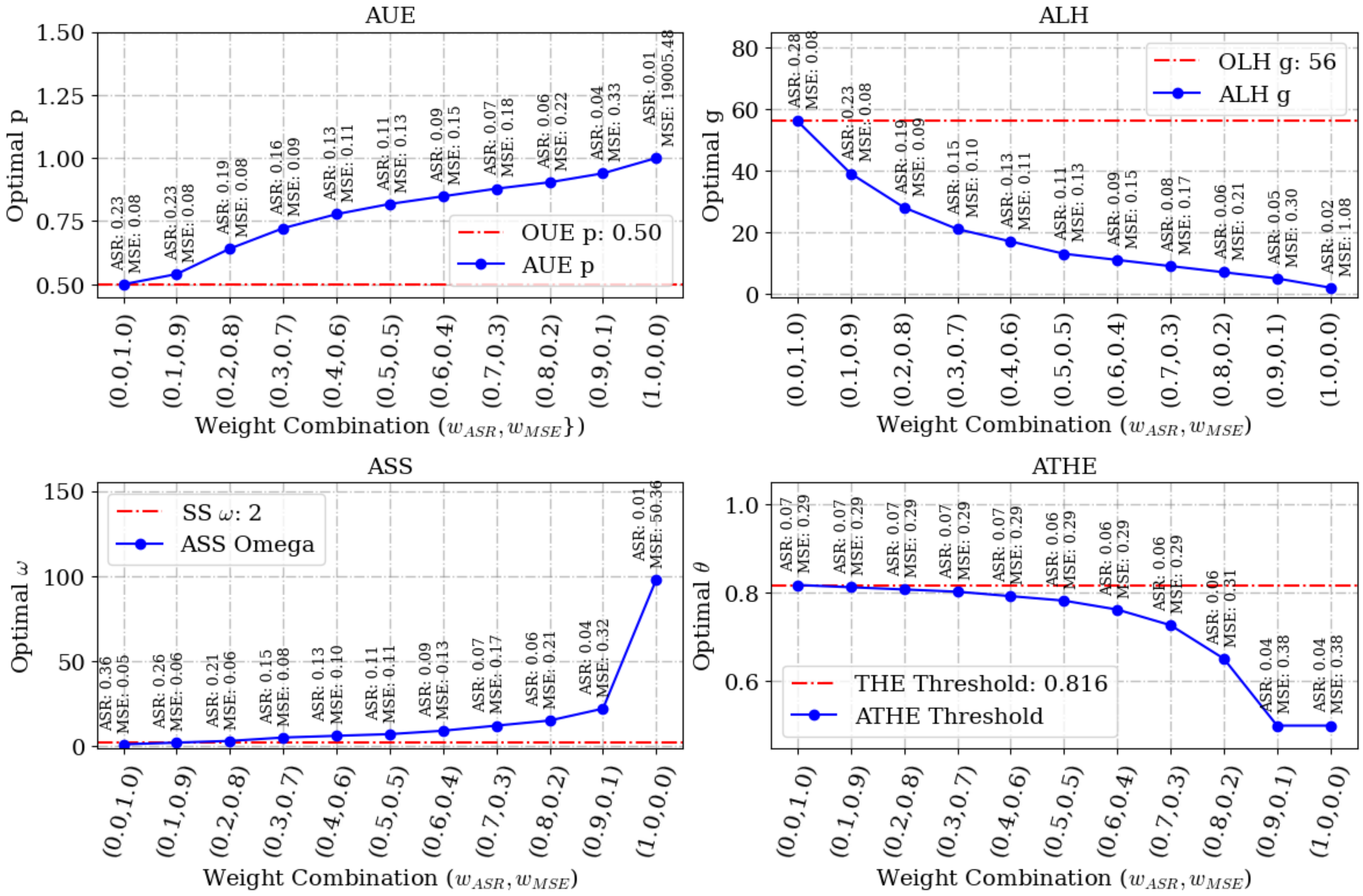
Impact of Weights on Parameter Selection



Ablation: Multi-Objective Rules